```r
library(quantreg)
```

```
## Loading required package:  SparseM
```

```r
set.seed(42)

n<-100
x <- (1:n)/10

e1 <- 1+2*x+(1/2+x)*rnorm(n)
e2 <- 5 * (rexp(n, 1) - log(2))
e3 <- rt(n, 2)

y1 <- 1+2*x + e1
y2 <- 1+2*x + e2
y3 <- 1+2*x + e3
```

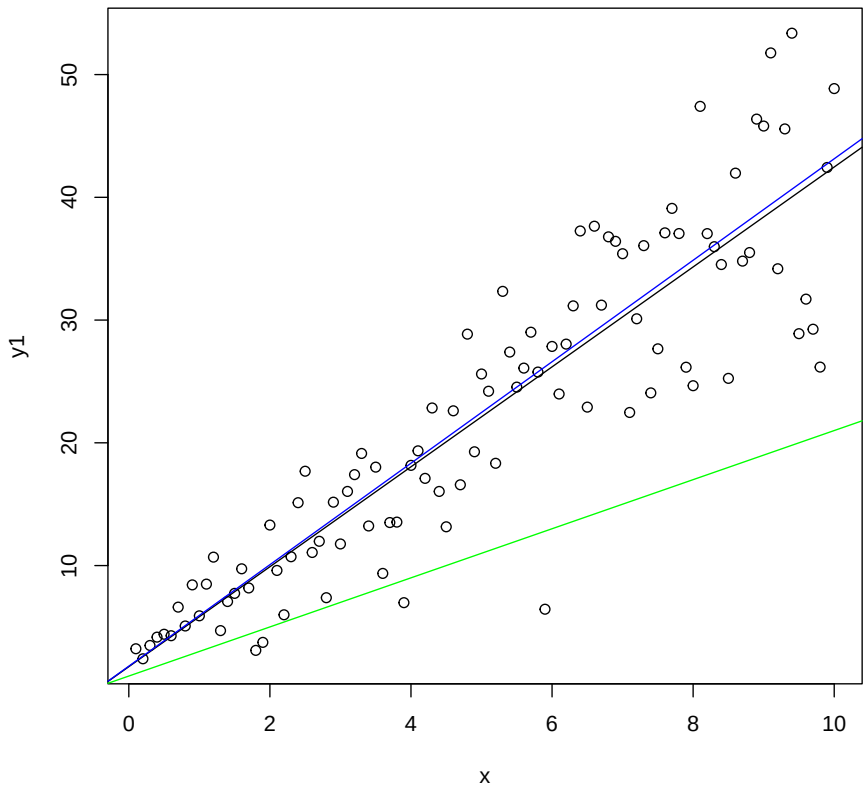
```r
model1 <- lm(y1 ~ x)
model2 <- lm(y2 ~ x)
model3 <- lm(y3 ~ x)
plot(x,y1)
abline(coef(model1))


tau <- c(0.05, 0.25, 0.5, 0.75, 0.95)

library(cqrReg)
```

```
## Loading required package:  Rcpp
```

```r
model.mediana <- rq(y1~x, tau=0.5)
abline(coef(model.mediana), col='blue')
abline(1,2,col='green')
```


y1
x
10
20
30
40
50
0
2
4
6
8
10